# 2.REVIEW OF DIGITAL COMPONENTS




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# REGISTERS

- A register is a group of flip-flops capable of storing one bit of information.
- An n-bit register has a group of n flip-flops and is capable of storing any binary information of n bits.
- In addition to flip-flops, registers can have combinational gates that perform certain data-processing tasks.
- The flip-flops hold binary information and the gates control when and how new information is transferred into the register.
- The simplest possible register is one that consists of only flip-flops without any external gates.
- Figure 2-1 shows such a register constructed with four D-type flip-flops and a common clock-pulse input.

# REGISTERS(cont'd...)

- The common clock input triggers all flip-flops on the rising edge of each pulse, and the binary data available at the four inputs are transferred into the 4-bit register.
- The clear input *is* useful for clearing *the* register to all *0's* prior to its clocked operation.
- Note that *the clock signal enables the D input but that the clear input is independent of the clock.*

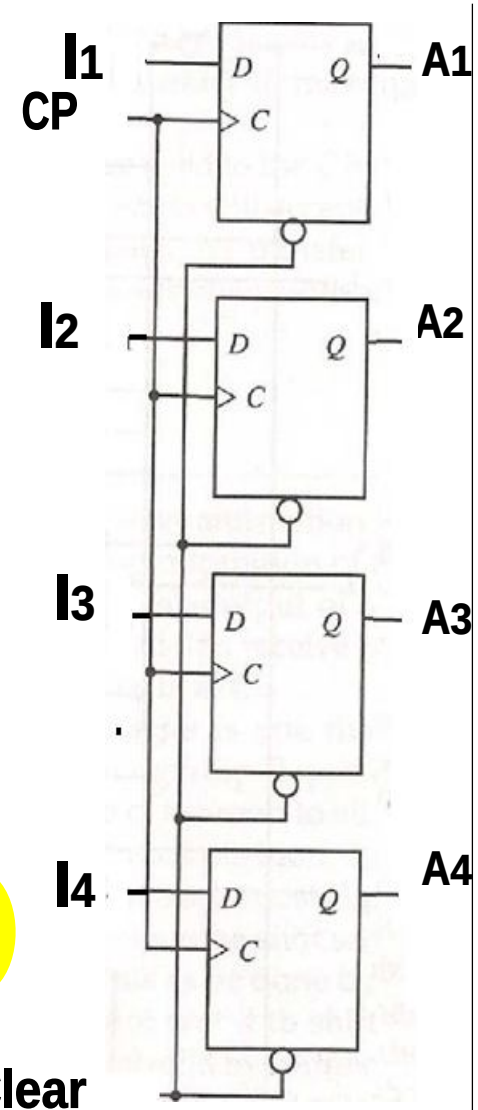



Fig. 2-1 4-bits register

# Register load

- The transfer of new information into a register is referred to as a loading the register.
- If all the bits of the register are loaded simultaneously with a single clock pulse, we say that the loading is done in parallel.
- A clock transition applied to the Clock inputs of the register of Fig. 2-1 will load all four inputs $I_1$ through $I_4$ in parallel.
- In this configuration, the clock must be inhibited(prevented) from the circuit if the content of the register must be left unchanged.
- In other words, the Clk input act as an enable signal that controls the loading of new information into the register

# Register with Parallel Load

- Most digital systems have a master clock generator that supplies a continuous train of clock pulses.
- A 4-bit register with a load control input using D FFs is shown in Fig. 2-2.
- The Clock inputs receive clock pulses at all times.
- The purpose of the buffer gate in the clock input is to reduce the power requirement from master-clock generator(to increase "fan-out").

# Register with Parallel Load(Cont'd...)

- In this register, the operation of the register is controlled by the load input.
- Load Input
  - 1 : Four input transfer
  - 0 : Input inhibited, Feedback from output to input( *no change*)
- Note that the clock pulses are applied to the Clock input at all times.

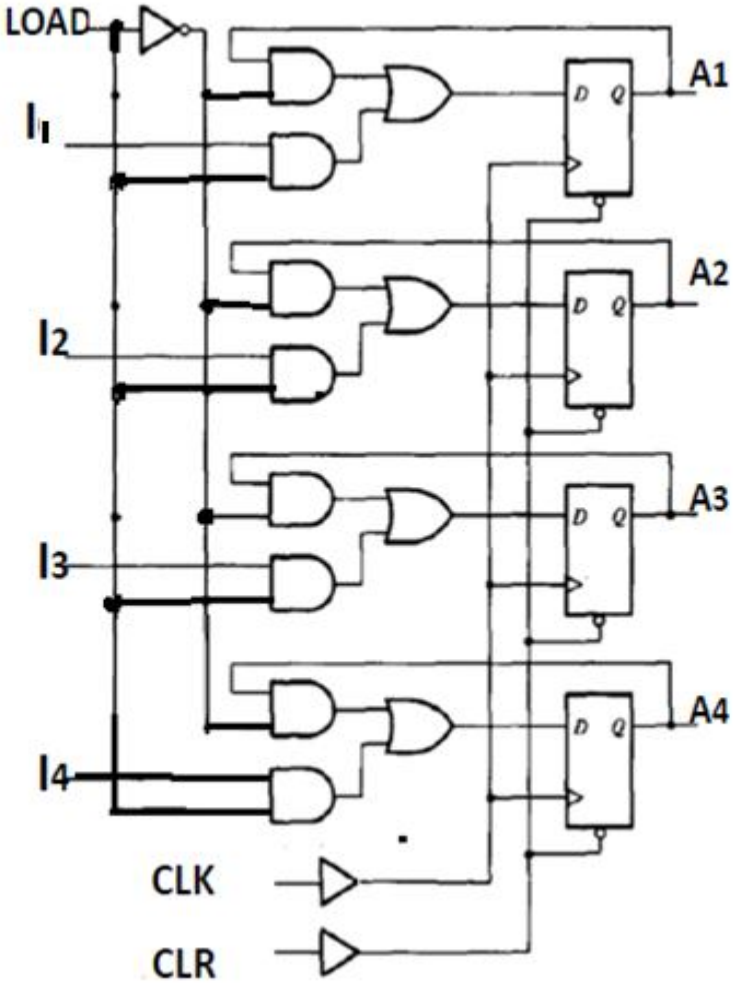



Fig. 2-2 Register with parallel Load using *D* flip-flops

# Shift Registers

- A register capable of shifting its binary information in one or both directions is called a shift register.
- Shift registers are constructed by connecting flip-flops in cascade, where the output of one flip-flop is connected to the input of the next flip-flop.
- All flip-flops receive common clock pulses that initiate the shift from one stage to the next.
- The simplest possible shift register is one that uses only flip-flops, as shown in Fig. 2-3.

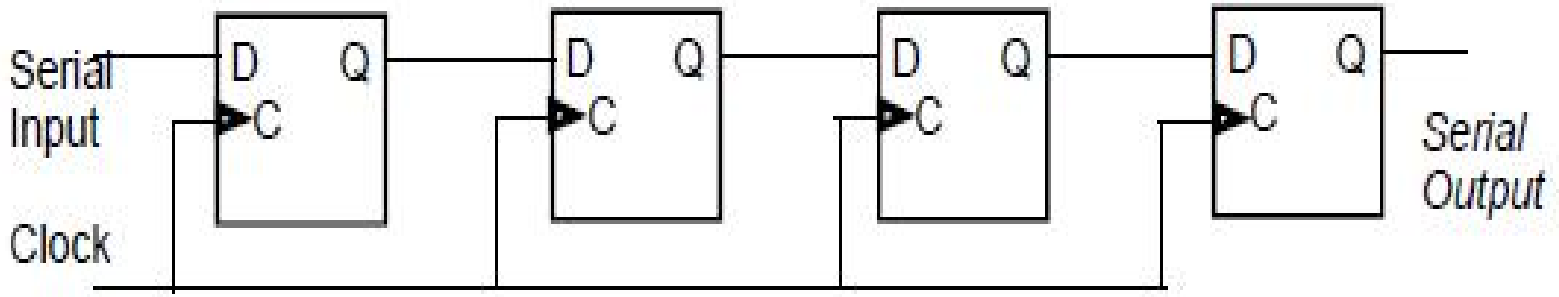



Fig. 2-3. 4-bit shift register

# Cont'd

- **The *serial input* determines what goes into the leftmost position during the shift and**
- **The *serial output* is taken from the output of the rightmost flip-flop.**
- ***The register in Fig. 2.3 shifts its contents with every clock pulse.***
- **If we want to control the shift, we must control the Clock input of the register.**
- **If, however, the shift register in Fig. 2.3 is used, the shift can easily be controlled by means of an external AND gate, as shown in what follows.**

# Serial Transfer(Cont'd)

- A digital system is said to operate in a serial mode when information is transferred and manipulated one bit at a time.
- The information is transferred one bit at a time by shifting the bits out of the source register into the destination register.
- The serial transfer from register A to register B is done with shift registers, as shown in the block diagram of Fig. 2.4(a).
- The serial output (SO) of register A goes to the serial input (SI) of register B.

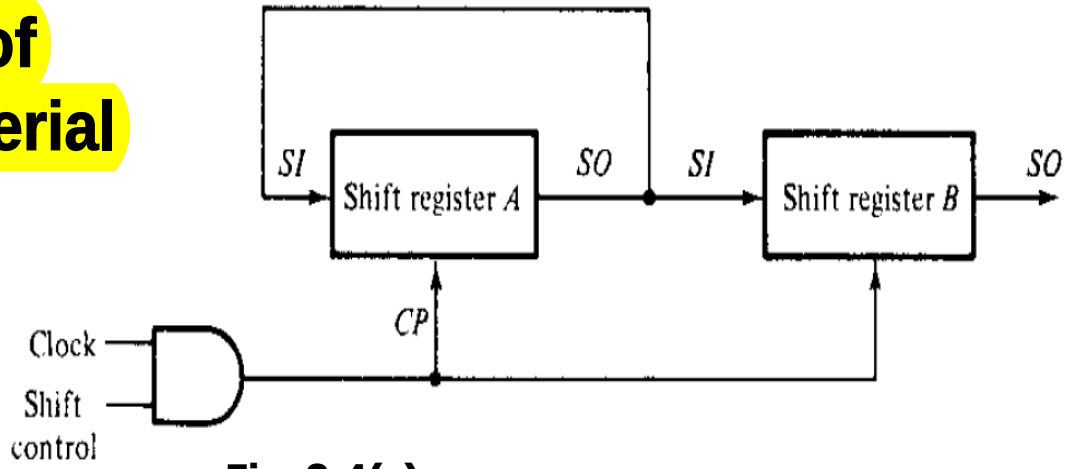



**Fig. 2.4(a).** (a) Block diagram

# Cont'd

- To prevent the loss of information stored in the source register, the A register is made to circulate by connecting the serial output to its serial input terminal.
- The initial content of register B is shifted out through its serial output and is lost unless it is transferred to a third shift register.
- The shift control input determines when and by how many times the registers are shifted.
- This is done by the AND gate that allows clock pulses to pass into the CP terminals only when the shift control is 1

# Cont'd

- **E.g. Suppose the shift registers have four bits each.**
- **The control unit that supervises the transfer must enables the shift registers, through the shift-control signal, for a fixed time duration equal to four clock pulses. This is shown in the timing diagram of Fig. 2.4 (b).**
- **For n-bits register, n-clock pulses are required to transfer all n-bits.**

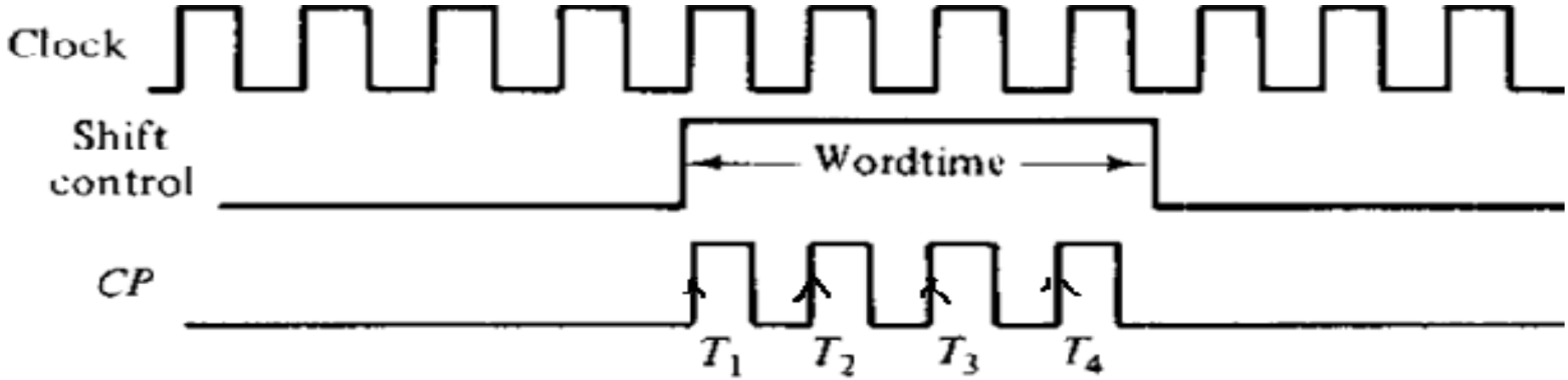



**Fig. 2.4 (b).timing diagram**

# Cont'd

- The shift-control signal is synchronized with the clock and changes value just after the positive edge of a clock pulse.
- The next four clock pulses find the shift-control signal in the 1 state, so the output of the AND gate connected to the CP terminals produces four pulses, T1. T2, T3, and T4.
- E.g. : Assume that the binary content of A before the shift is 1011 and that of B, 0010. after the fourth shift, what would be the contents of both register A and register B?

| Timing Pulse | Shift Register A | | | | Shift Register B | | | |
|---|---|---|---|---|---|---|---|---|
| Initial value | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| After $T_1$ | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| After $T_2$ | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| After $T_3$ | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 |
| After $T_4$ | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 |

- After the fourth shift, both registers A and B have the value 1011.
- Thus, the content of A is transferred into B, while the content of A remains unchanged.

# Cont'd

- The difference between serial and parallel modes of operation should be apparent from this example.

| Serial transfer mode | Parallel transfer mode |
|---|---|
| ➢ The operation is slower | ➢ The operation is faster |
| ➢ Only one bit is transferred during one clock-pulse | ➢ All bits transferred simultaneously during one clock-pulse |
| ➢ For n-bits register, n-clock pulses are required to transfer all n-bits. | ➢ For n-bits register, 1-clock pulse is required to transfer all n-bits. |
| ➢ Require less hardware | ➢ Require more hardware |

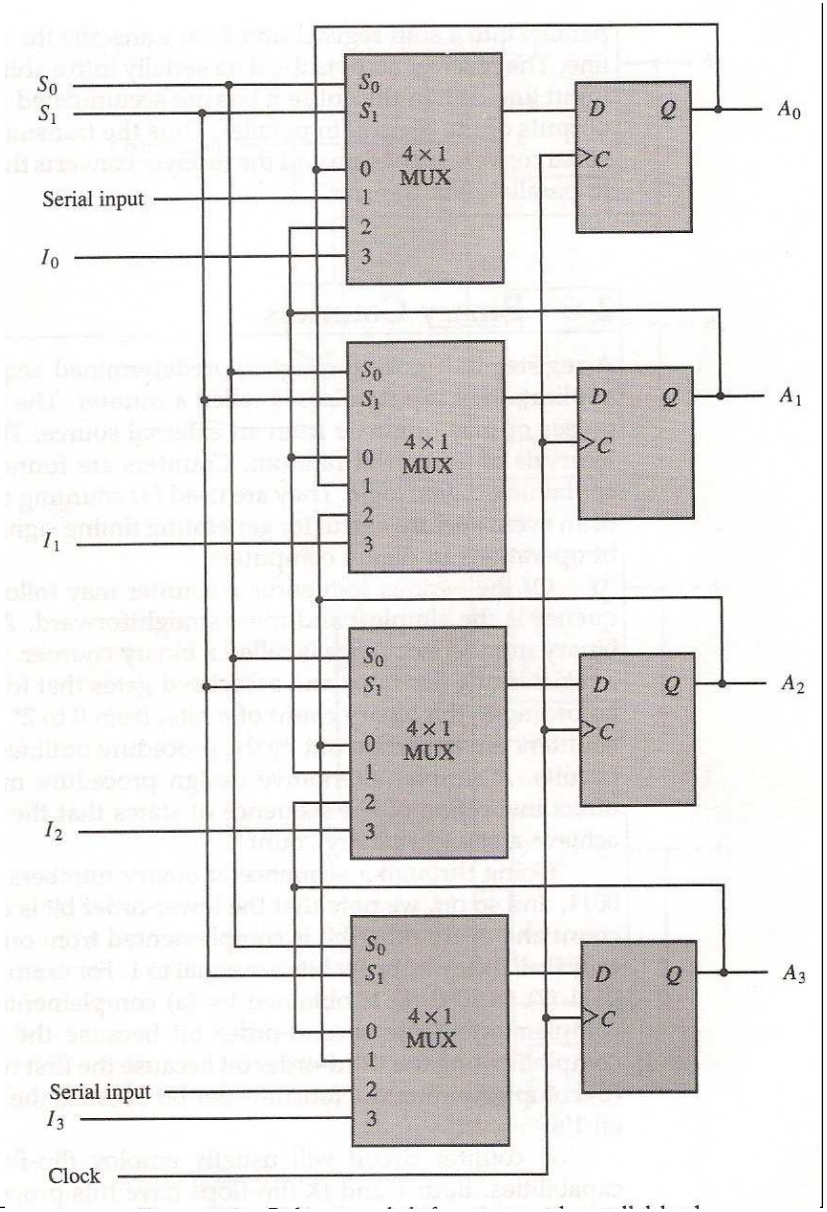
# Bidirectional Shift Register with Parallel Load

- A register capable of shifting in one direction only is called a unidirectional shift register
- A register that can shift in both directions is called a bidirectional shift register
- The most general shift register has all the capabilities listed below:
  - An input for clock pulses to synchronize all operations.
  - A shift-right operation and a serial input line associated with the shift-right.
  - A shift-left operation and a serial input line associated with the shift-left.
  - A parallel load operation and n input lines associated with the parallel transfer.
  - n parallel output lines.
  - A control state that leaves the information in the register unchanged even though clock pulses are applied continuously.

# Cont'd

- A 4-bit bidirectional shift register with parallel load is shown in Fig. 2-9.
- Each stage consists of a *D flip-flop and a 4 x 1 multiplexer.*
- *The two selection* inputs S1 and *So select one of the multiplexer data inputs for the D flip-flop.*
- The selection lines control the *mode of operation* of the register according to the function table shown in Table 2-1.

| Mode | | Operation |
|---|---|---|
| S1 | S0 | |
| 0 | 0 | No chage |
| 0 | 1 | Shift right(down) |
| 1 | 0 | shift left(up) |
| 1 | 1 | Parallel load |



# Memory Unit

- A *Memory unit* is a collection of storage cells together with associated circuits needed to transfer information in and out of storage.
- Memory stores binary information in groups of bits called *words.*
- *Word* is a group of binary information that is processed in one simultaneous operation
- A group of eight bits is called *a byte*. Most systems use *memory words* that are a multiple of eight.
- Thus, a *16-bit word* contains *two bytes*. *A 32-bit word* contains *four bytes*, and so on.
- The number of *words* it contains and the number of *bits* in each word define *the internal structure of a memory unit*.
- Special input lines called *address lines* select one particular word.

# Cont'd

- Each word in memory is assigned a *unique address* ranging from *0 to* $2^{k-1}$, where k is the number of address lines.
- Applying the *k-bit binary address* to the *address lines* does the selection of a *specific word* in memory.
- A decoder inside the memory accepts the address and opens the paths needed to select the bits of the specified word.
- Computer memories range from *1024 words*, requiring an *address of 10 bits*, to $2^{32}$ *words*, requiring *32 address bits*.
- It is customary to refer to the number of words in a memory with one of the letters K (kilo), M (mega), G (giga), where K is equal to $2^{10}$, M is equal to $2^{20}$, G is equal to $2^{30}$.
- Thus, 64K = $2^{16}$, 2M = $2^{21}$, and 4G = $2^{32}$.

# Cont'd

- **Two major types of memory are used in computer systems:**

- **Random-Access Memory (RAM)**
  - **The memory cells can be accessed for information transfer from any desired random location**
  - **Communication between a memory and its environment is achieved through *data input and output lines, address selection lines, and control lines* : *Fig. 2-6***

- **The two operations that a random-access memory can perform are the *write and read operations***
- ***Write: transfer of data into a memory***
- ***Read: transfer of data out of a memory***

# Cont'd

- **Memory Write**
  - Place the binary address of word on address selection lines
  - Place the data bits on the data input lines and
  - The write control line is activated.
- **Memory Read**
  - Place the binary address of word on address selection lines
  - The read control line is activated
  - The memory unit then places the desired data onto the data output lines.

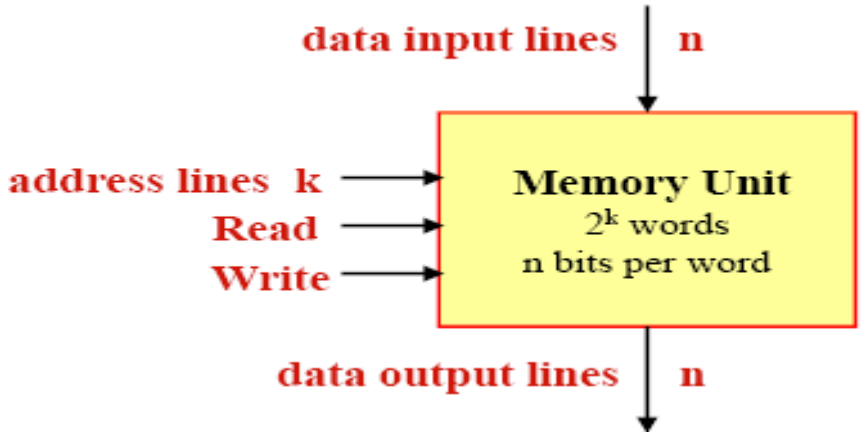



*Fig. 2-6 Block diagram of RAM*

# Cont'd

- **Read-Only Memory (ROM)**
  - The read-only memory can only perform the read operation, *it* does not have a write capability.
  - This implies that the binary information stored in a ROM is made permanent during the hardware production of the unit and can not altered by writing different word into it.
  - The read operation on a ROM is identical to that of a RAM with the exception being that there is no need for a read control line

- Once the pattern is established, it stays within the unit even when power is turned off and on again.
- An mxn ROM has k address input lines to select one of 2k = m words of memory, and n output lines(n bits word)

